

A Machine Learning Framework for Measles Detection and Localized Outbreak Risk Prediction in Bangladesh

Tahmid Tamjid Turag

Department of Computer Science
American International University-Bangladesh (AIUB)
23-50625-1@student.aiub.edu

Md. Uswataul Hasan Nibir

Department of Computer Science
American International University-Bangladesh (AIUB)
23-50638-1@student.aiub.edu

Abstract—Measles is a highly contagious, airborne viral disease, and a major public health problem in Bangladesh in high density and resource poor settings. Gold-standard laboratory diagnostics such as RT-PCR and IgM serology provide definitive confirmation, but delays in central lab processing can severely hamper rapid frontline outbreak containment. This paper fills this important gap by proposing a non-invasive point-of-care machine learning framework to screen measles cases and predict localized outbreak risks based on routinely collected clinical, demographic, vaccination, and spatiotemporal surveillance data. We utilize a simple-average ensemble of three tree-based gradient boosting algorithms, namely XGBoost, CatBoost, and LightGBM. The proposed case-prediction ensemble achieved ROC-AUC of 0.8489, PR-AUC of 0.9024, overall accuracy of 0.7928 and sensitivity of 0.8830 when evaluated on an untouched test set of 15 selected features. The framework achieved an Out-of-Fold (OOF) ROC-AUC of 0.8956 for outbreak prediction at the Upazila/Municipality/City Corporation (UPZMUNCC) level with a cell prevalence of 7.3%. Moreover, SHapley Additive exPlanations (SHAP) are used to interpret the model predictions. The main driver features are epidemiological timing (epiweek), measles vaccination doses (doses_mcv), child age and duration of fever. This framework provides immediate, interpretable risk assessments before laboratory confirmation, allowing frontline health officers to conduct timely contact tracing and targeted ring vaccination within 24 hours to interrupt localized transmission of the disease

Index Terms—Measles, machine learning, point-of-care screening, outbreak prediction, ensemble learning, SHAP, Bangladesh

I. INTRODUCTION

Measles or "Ham" is a virus that is airborne and very contagious from person-to-person via sneezing and coughing. Symptoms usually include high fever, cough, running nose and red eyes, followed by small white spots in the mouth (Koplik spots) and a red rash which begins on the face and then descends down the body. In addition to these symptoms, measles can temporarily weaken the immune system, resulting in serious complications like pneumonia, diarrhea, brain swelling or even blindness. In a densely populated country such as Bangladesh, measles is continuing to be a significant public health issue in refugee camps in Cox's Bazar and urban slums in Dhaka. A large number of reported cases are in young

children and during outbreaks, pediatric wards and hospitals are overwhelmed with the need to provide isolation beds, oxygen, medicines, such as Vitamin A, that can save lives — fortunately, measles is almost 100% preventable. Bangladesh, in turn, has an Expanded Programme on Immunisation (EPI) which provides two doses of Measles-Rubella (MR) vaccine at 9 and 15 months of age. Where outbreaks occur, specific mass immunization campaigns are conducted to immunize unvaccinated children, to fill the vaccine gaps and to ensure community safety.

Laboratory confirmation of measles is the clinical gold standard, and involves IgM serology or RT-PCR, which requires days to return results from central laboratories [1]–[4]. Since then, machine learning has shown potential to predict cases of measles and forecast outbreaks [5]–[9] but most existing studies have been conducted in high-income nations and/or at the international level, while little work has been done on Bangladesh specific surveillance data, vaccination patterns, and geography.

Early diagnosis of measles is a race in Bangladesh. Presently, blood samples are taken to the central labs of Dhaka from rural Upazilas and the lab takes days or even weeks to provide official confirmation. If the child remains undiagnosed, they can be unwittingly spreading the virus to their peers and neighbors during this critical window creating a widespread outbreak.

This project aims to change this by bringing the screening tool to the frontline health care worker, making it an instant tool and intelligent. Rather than relying on delayed lab results, our machine learning model uses basic information that we already collect when a child comes to us for a routine visit to a clinic — including the date the child started showing symptoms, how long symptoms have been present, a child's vaccination record and local case trends — to identify measles at the point of care. For this purpose, we apply ensemble of XGBoost, CatBoost and LightGBM, where predictions are made by all the models and then combined together. For this purpose, we apply ensemble of XGBoost, CatBoost and LightGBM, where predictions are made by all the models and then combined together. As the task is to classify whether a

case falls into the measles or not measles class, the underlying models give out the probability by using a sigmoid function and the final prediction of the ensemble is the averages of the probabilities predicted by the three models.

The important thing to note is that this system isn't merely spitting out a random percentage. In the case of zero-dose vaccination status, symptom timing or location, the model using the explainable AI technique SHAP (SHapley Additive exPlanations) clearly explains to health officers the reason for the case being flagged. This allows local health teams to act decisively, and to trace contacts and vaccinate surrounding children within 24 hours to prevent an outbreak from spreading, before a lab report returns.

II. BACKGROUND STUDY

The spread of measles, the risk of outbreaks and clinical diagnosis have been explored in a variety of data settings using different computational and predictive models [1]–[14]. Because of their good performance on tabular epidemiological records, the tree-based ensemble classifiers are also used as primary statistical baselines, such as Extreme Gradient Boosting (XGBoost) and Random Forests [8], [9]. The conventional mechanistic models, such as the Time-Series Susceptible Infected-Recovered (TSIR) model, are still widely used in tracking diseases [5], [7]. In recent studies, a combination of mechanistic rules and neural networks has been introduced, leading to hybrid systems like Spatial-Feature Neural Networks (SFNN) and TSIR-Augmented Physics Informed Neural Networks (TSIR-PINN) [5]. In the diagnostic field, the deep learning approach such as Vision Transformers (DeiT) and MobileNetV2 with Gradient Boosting Classifier is used for the detection of rashes in the skin images [12], [13]. Other commonly used models include statistical LASSO regression models, Naive Bayes and Support Vector Machines (SVM), Multi Layer Perceptrons (MLP) [7], [8]. Moreover, molecular diagnostic techniques employ Receiver Operating Characteristic (ROC) curve models and linear regression of real-time RT-PCR viral loads [1], [2], [4].

The primary aim of these studies is to allow early identification, precise estimation of outbreak hazard, and increase the precision of outbreak diagnosis, to support public health interventions [6], [8], [11]. The goal of a predictive modeling effort is to predict trends in spatiotemporal transmission, estimate county-level risk, and detect super-spreading events prior to large-scale outbreaks [3], [6], [9]. For clinical diagnosis, researchers develop automated point-of-care devices that assist health care workers in the detection of skin rashes with various skin color types [12], [13]. Other investigations are directed at molecular surveillance and environmental testing, which include testing assays that detect viral shedding in communities in real-time RT-PCR and wastewater testing [1], [10], [14]. Additionally, researchers make efforts to close the gap between black-box statistical models and epidemiology by integrating both methods with physical disease mechanisms [5], [7]. In summery these goals aim at the goal of improving early

warning systems, strengthening outbreak control measures and reducing reporting delays [6], [9], [10].

Although there have been some advances, a few limitations are encountered in the current literature [5], [6], [8]. It is difficult to capture the complexity and nonlinearity of spatial interactions between regions while retaining interpretability in a traditional epidemiological framework [5]. In contrast to machine learning models, often, they are lacking in fine-grained spatial and mobility predictors and primarily using static geographical averages that fail to capture the localized dynamics of outbreaks [6], [9]. The limitations of computer vision systems for rash detection include less representative training sets, with insufficient amounts of dark skin tones or contextual clinical information such as fever durations [12], [13]. Wastewater surveillance is difficult to distinguish wild-type measles outbreaks from vaccine strain shedding, which is an expected phenomenon of live-attenuated vaccines [14]. Lastly, many tabular machine learning models have been trained with highly imbalanced data and missing data in routine clinical surveillance data, making them less reliable when deployed in real-life, low-resource environments [8].

III. METHODOLOGY

The methodology suggested is systematic and takes data collection and analysis, prediction, and presentation of the results. Data analysis consists of data preprocessing, feature engineering, handling of class imbalance, development of an ensemble model, explainability analysis and performance evaluation.

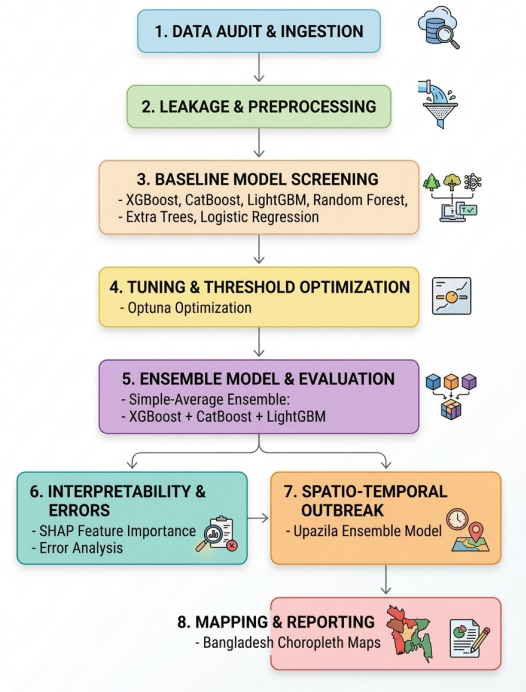


Fig. 1. Proposed methodological workflow for baseline screening, ensemble model evaluation, interpretability, and spatio-temporal outbreak mapping.

A. Dataset and Data Preprocessing

The data is from the MR Data sheet and has 30,252 records and 58 variables. The target variable was created by those who were diagnosed with measles were assigned the positive class (“2-Laboratory Confirmed Measles”), and those who were not diagnosed with measles were assigned the negative class (“6-Discarded” or “4-Laboratory Confirmed Rubella”). Records were identified with an outcome of “pending” and were not included in supervised training because the outcome of the disease was not yet available for those records. Of the final labeled data set, 5,751 cases were confirmed and 3,126 were excluded as negative cases, with 21,375 pending cases. Laboratory and post-diagnosis variables were not included in the model inputs to avoid leakage of information and to facilitate early prediction at presentation. These were MeaslesIgM, RubellaIgM, serology, urine / swab results, genotypes, Comment and CASE_INV related fields. Other variables that could be collected at the time of presentation, such as age, sex, vaccination history, timing of fever, timing of rash, travel history and division, district and upazila/municipality/city corporation were retained. Data on high cardinality location identifiers (such as village/mahal, union/ward, and facility) were excluded because they have little clinical signal and are almost always present for every case.

B. Feature Engineering and Class Imbalance Handling

For the feature ranking, we used the Random Forest importance and mutual information to select informative features. Epidemiological timing, geographic information, age, duration of fever, duration of rash, vaccination status, vaccination doses, sex, duration of investigation and clinical indicators were examined. Stratified cross-validation with XGBoost was used to evaluate several features subsets. The 15-feature subset was chosen with an AUC of 0.8411 ± 0.009 for cross-validation and was then used to compare the main models. The 15-feature subset was selected and had an AUC of 0.8411 ± 0.009 for cross-validation, and was then used for comparing the main models. The class distribution for the labelled data was imbalanced, with 5812 cases of measles and 3182 cases of non-measles. Five-fold cross-validation was used to assess various class-weighting schemes. The modelling process thus took class imbalance into account by making a comparison based on validation rather than a resampling method unsupported. The selected features were used to design the final ensemble, which was validated by the procedure described in the next subsection.

C. Proposed Ensemble Machine Learning Framework

The proposed framework uses three gradient-boosting classifiers: XGBoost, CatBoost, and LightGBM. Each model is trained independently using the selected features and produces a probability of measles for each case. The final ensemble probability is calculated using a simple average of the three model probabilities:

$$P_{\text{ensemble}} = \frac{P_{\text{XGB}} + P_{\text{CatBoost}} + P_{\text{LightGBM}}}{3}. \quad (1)$$

TABLE I
TEST-SET PERFORMANCE OF THE ENSEMBLE MODEL

Metric	Value
Accuracy	0.7928
ROC-AUC	0.8489
PR-AUC	0.9024
Sensitivity	0.8830
Specificity	0.6269
Precision	0.8132
F1-score	0.8467

D. Model Explainability and Performance Evaluation

To explain the predictions of the trained model and understand which features are most influential for the prediction, the model predictions were interpreted using SHAP (SHapley Additive exPlanations). Epidemiological, clinical, vaccination, demographic and geographic factors were analyzed. Fever duration, age, geographical variables, epiweek and measles vaccination doses had the highest mean absolute SHAP values. The status of vaccination, timing of symptoms and geographic information were also explored to enhance the interpretability of the model predictions. The accuracy, ROC-AUC, PR-AUC, sensitivity, specificity, precision, F1 score, Matthews correlation coefficient (MCC), negative predictive value (NPV), false-positive rate (FPR), and false-negative rate (FNR) were used to assess model performance, along with the Brier score. Apart from the primary test evaluation, error analysis was carried out to look at false positive and false negative cases, five random-seed stratified splits performed for a robustness check. These evaluations were used for the assessment of predictive performance and of the consistency of the proposed approach.

IV. RESULT AND DISCUSSION

A. Overall Performance

The proposed framework is a comparison of ensemble learning for measles case and outbreak level prediction. The final model for the case prediction is a simple average ensemble from XGBoost, CatBoost, and LightGBM. The validation ROC-AUC values for their models were 0.8483, 0.8512, and 0.8389 respectively with the ensemble at 0.8529. The threshold was set at 0.48 with the maximum validation accuracy to achieve an accuracy of 0.7845.

The ensemble performance on the unseen test set was 0.7928 accuracy, 0.8489 ROC-AUC and 0.9024 PR-AUC (Table II). Sensitivity, F1, MCC, precision, NPV, specificity, balanced accuracy, and Brier score were 0.8830, 0.8467, 0.5332, 0.8132, 0.7443, 0.6269, 0.7549, and 0.1474, respectively. The confusion matrix had 294 TN, 175 FP, 101 FN, and 762 TP cases (Fig. 3) and a balance between detection of cases in the positive category and false-positive control in the negative category.

Ensemble Confusion Matrix (thr=0.48)

True label	Not Measles	294	175
	Measles	101	762
		Not Measles	Measles
		Predicted label	

Fig. 2. Confusion matrix of the final simple-average XGBoost–CatBoost–LightGBM ensemble on the untouched measles case-prediction test set.

TABLE II
PERFORMANCE OF THE FINAL SIMPLE-AVERAGE ENSEMBLE FOR MEASLES CASE PREDICTION ON THE UNTOUCHED TEST SET.

Metric	Value
Accuracy	0.7928
ROC-AUC	0.8489
PR-AUC	0.9024
Sensitivity	0.8830
Specificity	0.6269
Precision	0.8132
NPV	0.7443
F1-score	0.8467
Balanced Accuracy	0.7549
MCC	0.5332
Brier Score	0.1474

Five random-seed based stratified splits were used for assessing the stability of the underlying XGBoost pipeline. The analysis is for the pipeline that uses XGBoost, not the ensemble. ROC-AUC ranged from 0.829 to 0.853, PR-AUC from 0.892 to 0.915, sensitivity from 0.738 to 0.900, and specificity from 0.544 to 0.795. Mean accuracy, ROC-AUC, PR-AUC, precision, F1, and MCC were 0.7688 ± 0.0087 , 0.8407 ± 0.0107 , 0.9009 ± 0.0089 , 0.8197 ± 0.0326 , 0.8222 ± 0.0166 , and 0.4953 ± 0.0138 , respectively.

Feature-subset experiments with XGBoost revealed that Model_C_top15 had an average ROC-AUC of 0.8411 ± 0.009 and a PR-AUC of 0.9025 with 15 features. The original subset and clinical subset had lower values, and Model_F and Model_B_top20 had slightly higher values. Model_C_top15 was chosen to be the final case-prediction model.

Outbreaks were defined as two or more laboratory confirmed measles cases in one unit week when making predictions at the outbreak level. The division, district, and UPZMUNCC levels were analyzed, and the simple-average XGBoost-CatBoost-LightGBM ensemble was the primary

TABLE III
FEATURE-SUBSET EXPERIMENTS USING XGBOOST WITH STRATIFIED CROSS-VALIDATION.

Subset	Features	ROC-AUC	PR-AUC
Model F	21	0.8425 ± 0.009	0.9045
Model B top20	20	0.8419 ± 0.010	0.9035
Model C top15	15	0.8411 ± 0.009	0.9025
Model D top10	10	0.8388 ± 0.009	0.9018
Model A original	13	0.7392 ± 0.011	0.8108
Model E clinical	11	0.7184 ± 0.011	0.7973

TABLE IV
OOF PERFORMANCE OF THE FINAL UPZMUNCC OUTBREAK ENSEMBLE.

Metric	Value
Units	461
Unit-week cells	10,142
Outbreak cells	743 (7.3%)
Scale positive weight	12.65
Threshold	0.43
Accuracy	0.9268
ROC-AUC	0.8956
PR-AUC	0.5175
Sensitivity	0.4980
Specificity	0.9607
F1-score	0.4993
MCC	0.4599
TN	9,030
FP	369
FN	373
TP	370

model at the UPZMUNCC level. This level contained 461 units, 10,142 unit-week cells, and 743 outbreak cells (7.3%). The `scale_pos_weight` used was 12.65 and the threshold was set to 0.43, based on the OOF predictions and the maximum F1.

The OOF accuracy of UPZMUNCC ensemble was 0.9268, ROC-AUC was 0.8956, and PR-AUC was 0.5175. Sensitivity and specificity were 0.4980 and 0.9607, while F1 and MCC were 0.4993 and 0.4599. The OOF confusion matrix contained 9,030 TN, 369 FP, 373 FN, and 370 TP. If there is an outbreak, as there was in this case, accuracy is not enough to judge performance.

B. Comparison with Individual Classifiers

Each of the classifiers was tested for the selected features using 5-fold cross validation. As shown in Table V, XGBoost, CatBoost, and LightGBM achieved ROC-AUC values of 0.8422, 0.8414, and 0.8387, with PR-AUC values of 0.9034, 0.9025, and 0.9028, respectively. Other classifiers yielded lower values and HistGradientBoosting was not included as it is not compatible with sparse-data/dense-data.

The validation ROC-AUC of the final simple-average ensemble is 0.8529, while XGBoost, CatBoost and LightGBM achieved 0.8483, 0.8512 and 0.8389, respectively. For the untouched test set, XGBoost reached an accuracy of 0.7793 and ROC-AUC of 0.8499 at threshold 0.38, with 228 false positives and 66 false negatives. The ensemble gave the best accuracy of 0.7928 and the highest ROC-AUC of 0.8489 at threshold 0.48, with 175 false positives and 101 false

TABLE V
FIVE-FOLD CROSS-VALIDATION COMPARISON OF INDIVIDUAL CLASSIFIERS FOR MEASLES CASE PREDICTION.

Model	ROC-AUC	PR-AUC	Accuracy	Sensitivity	Specificity	F1	MCC
XGBoost	0.8422	0.9034	0.7793	0.8866	0.5819	0.8389	0.4993
CatBoost	0.8414	0.9025	0.7775	0.8913	0.5681	0.8385	0.4940
LightGBM	0.8387	0.9028	0.7732	0.8798	0.5771	0.8341	0.4856
RandomForest	0.8197	0.8782	0.7631	0.8228	0.6532	0.8182	0.4784
ExtraTrees	0.7733	0.8362	0.7426	0.8480	0.5486	0.8102	0.4168
LogisticRegression	0.7688	0.8105	0.7350	0.7727	0.6657	0.7908	0.4311
LinearSVC	0.7589	0.8043	0.7528	0.8851	0.5096	0.8227	0.4331

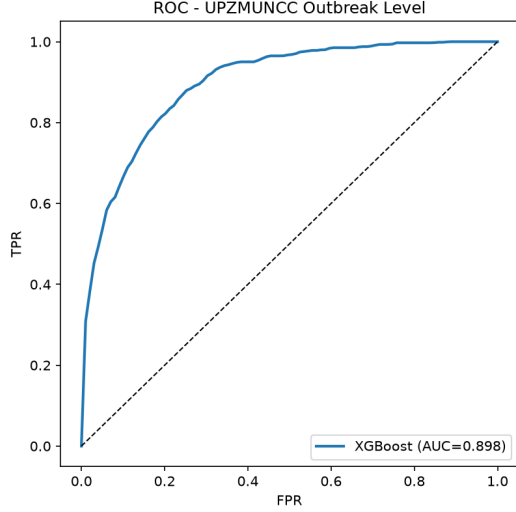


Fig. 3. ROC curve for the XGBoost-CatBoost-LightGBM outbreak ensemble at the UPZMUNCC level.

negatives, resulting in an increased accuracy of 0.0135 with a decreased number of false positive and an increased number of false negative.

Classifiers were tested at division, district and UPZMUNCC level for outbreak prediction and results were displayed in Tables VI–VII. ROC-AUC ranged from 0.8284–0.9191 at division level and 0.8579–0.9142 at district level; CatBoost achieved 0.9142 ROC-AUC and 0.8443 PR-AUC at district level. At UPZMUNCC level, ROC-AUC ranged from 0.7934–0.8982 and PR-AUC from 0.3363–0.5303 under the 7.3% outbreak-cell prevalence.

C. Explainability Analysis

SHAP values obtained through test-set predictions were used for explainability of the last model, XGBoost. As summarized in the Table VIII above, *epiweek* had the highest mean absolute SHAP value (1.1007), followed by *doses_mcv* (0.1735), *age* (0.1370), and *fever_duration* (0.1163). This was also influenced by geographic and clinical factors. These values are not causal, but outputs from a model.

TABLE VIII
TOP SHAP FEATURES FOR THE FINAL XGBOOST MODEL, RANKED BY MEAN ABSOLUTE SHAP VALUE.

Feature	Mean SHAP
<i>epiweek</i>	1.1007
<i>doses_mcv</i>	0.1735
<i>age</i>	0.1370
<i>fever_duration</i>	0.1163
<i>division_DHAKA</i>	0.1107
<i>ccc_YES</i>	0.0717
<i>rash_duration</i>	0.0585
<i>sex_FEMALE</i>	0.0572
<i>division_RANGPUR</i>	0.0378
<i>ccc_UNKNOWN</i>	0.0284
<i>district_CHITTAGONG</i>	0.0257
<i>division_SYLHET</i>	0.0248
<i>upzmunc_DHAKA NORTH CC.</i>	0.0218
<i>division_CHITTAGONG</i>	0.0213
<i>vax_status_VACCINATED</i>	0.0197

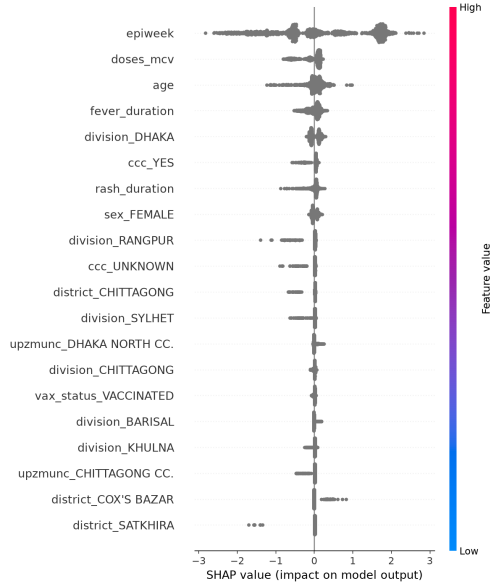


Fig. 4. SHAP beeswarm visualization for the final XGBoost model.

Typical predicted probabilities were a most confident prediction of Measles (0.984) and a least confident prediction of a true Measles (0.027) that was misclassified. The probability of the highest (Not Measles) case was 0.011 and the probability of a high risk false positive was 0.959.

The standalone XGBoost error analysis does not depict the final ensemble at threshold 0.48, but rather threshold 0.64. It

TABLE VI
INDIVIDUAL OUTBREAK-CLASSIFIER COMPARISON AT THE DIVISION LEVEL.

Model	ROC-AUC	PR-AUC	Accuracy	Sensitivity	Specificity	F1	MCC
LogisticRegression	0.9039	0.9549	0.8133	0.8593	0.7141	0.8616	0.5759
DecisionTree	0.8284	0.8751	0.8035	0.9111	0.5718	0.8630	0.5301
NaiveBayes	0.8829	0.9382	0.7628	0.6667	0.9679	0.7866	0.5993
RandomForest	0.9139	0.9597	0.8538	0.9111	0.7321	0.8950	0.6568
ExtraTrees	0.8894	0.9392	0.8185	0.8667	0.7167	0.8672	0.5814
HistGradientBoosting	0.9191	0.9626	0.8540	0.9111	0.7308	0.8944	0.6600
XGBoost	0.8915	0.9500	0.8182	0.8667	0.7141	0.8670	0.5798
LightGBM	0.8928	0.9509	0.8335	0.8889	0.7141	0.8795	0.6117
CatBoost	0.9079	0.9564	0.8338	0.8815	0.7333	0.8784	0.6175

TABLE VII
INDIVIDUAL OUTBREAK-CLASSIFIER COMPARISON AT THE UPZMUNCC LEVEL.

Model	ROC-AUC	PR-AUC	Accuracy	Sensitivity	Specificity	F1	MCC
Logistic Regression	0.8724	0.4831	0.9372	0.2154	0.9943	0.3337	0.3812
Decision Tree	0.8654	0.4429	0.9365	0.2423	0.9914	0.3580	0.3843
Naive Bayes	0.8554	0.4165	0.9265	0.3514	0.9720	0.4117	0.3822
Random Forest	0.8492	0.4433	0.9321	0.3164	0.9807	0.4035	0.3899
Extra Trees	0.7934	0.3363	0.9255	0.3204	0.9733	0.3840	0.3566
HistGradientBoosting	0.8873	0.5013	0.9382	0.3123	0.9877	0.4251	0.4301
XGBoost	0.8982	0.5303	0.9395	0.2908	0.9907	0.4121	0.4304
LightGBM	0.8940	0.5162	0.9389	0.2908	0.9901	0.4104	0.4262
CatBoost	0.8965	0.5235	0.9384	0.2760	0.9907	0.3959	0.4155

had 96 false positives, 226 false negatives, 637 true positives and 373 true negatives. 39 children were younger than 1 year and 31 were 1–5 years old who were false positive; 94 and 82 children, respectively, were false negative. In total, 90 false positives and 190 false negative cases occurred with CCC, and 87 and 148 cases had no MCV dose registered, respectively. The median durations of fever and rash for false positive and true negative were 6.0 and 4.0 days, and 5.0 and 3.0 days respectively.

Geographically, false positives were mostly located in the DHAKA (44) and CHITTAGONG (30) area, and false negatives in the DHAKA (93), CHITTAGONG (38) and MYMENSINGH (32) area. The following are descriptive test-set patterns and do not provide geographic causation.

The burden and outbreak-risk maps are complementary representations of the assessed 2026 results for spatial outbreak analysis. The mapping pipeline matched 64/64 districts for the both maps. The UPZMUNCC outbreak ensemble is not a multi-year forecast, but rather represents within-year status estimation.

The test ROC-AUC and PR-AUC were 0.8489 and 0.9024, respectively, and the sensitivity and specificity of the case-level ensemble were 0.8830 and 0.6269, respectively. The outbreak ensemble produced by the UPZMUNCC had an OOF ROC-AUC of 0.8956 and PR-AUC of 0.5175 respectively, with the sensitivity of 0.4980 and specificity of 0.9607 for the 7.3% outbreak-cell prevalence. Case-level metrics are from the untouched test set, whereas outbreak-level metrics are OOF results.

Bangladesh - district outbreak risk (2026)

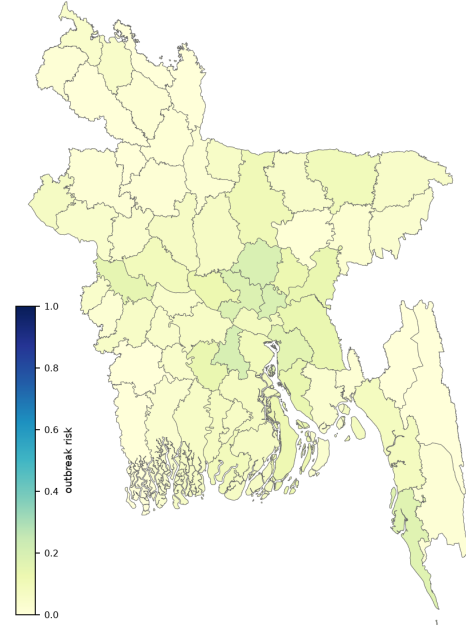


Fig. 5. Bangladesh district-level outbreak-risk visualization derived from the UPZMUNCC outbreak ensemble results

V. LIMITATIONS

There are a number of limitations in the proposed framework, particularly with respect to its predictive accuracy in frontline screening, but it shows promising results. The case-level ensemble has high sensitivity (0.8830), but low specificity (0.6269) with 175 false positives and 101 false

negatives; whereas the UPZMUNCC outbreak model has high specificity (0.9607), but low sensitivity (0.4980) under 7.3% prevalence. The process of routine surveillance may impact the reliability of its use in reality, due to the possibility of missing or imbalanced reporting, across upazilas and low-resource health centres. To boost the signal to noise ratio, missingness and reduce spatial aggregation, only high cardinality locations (villages/mahals and unions/wards) were included, thus limiting the ultra-local spatial representation. SHAP does not imply causal relationships, but rather provides explanations for the contributions of predictors like ‘epiweek’, ‘doses_mcv’, or duration of fever. Last, the spatial mapping does not make multi-year dynamic forecasts but rather within-year estimates of outbreak status, which reduces the extent of the projections.

VI. CONCLUSION

This study presents an explainable ensemble machine learning framework—combining XGBoost, CatBoost, and LightGBM—for rapid, point-of-care measles screening and spatial outbreak prediction in Bangladesh. Rather than waiting on laboratory confirmation, which can take several days, the model uses readily available demographic, temporal, geographic, and basic clinical data to flag likely cases early. It performs well across key metrics (ROC-AUC 0.8489, PR-AUC 0.9024, sensitivity 0.8830), and SHAP-based explainability makes its predictions interpretable, helping health workers see which factors—like zero-dose vaccination status or how long symptoms have persisted—are driving a given risk score. In practice, this means frontline officers in both rural upazilas and crowded urban areas can launch contact tracing and ring vaccination within 24 hours, containing outbreaks well before lab results come back. Going forward, we plan to incorporate real-time field data, improve sensitivity in highly imbalanced outbreak scenarios, and factor in population mobility to make early-warning systems even more robust in low-resource settings.

REFERENCES

- [1] J. P. Sevilla *et al.*, “Development of quantitative gene-specific real-time rt-pcr assays for the detection of measles virus in clinical specimens,” *Journal of Virological Methods*, vol. 132, no. 1–2, pp. 160–168, Mar. 2006.
- [2] Y. S. Lee *et al.*, “Development and evaluation of a new measles detection assay using real-time rt-pcr,” *International Journal of Molecular Sciences*, vol. 26, no. 5, p. 1801, Feb. 2025.
- [3] K. S. Tanaka *et al.*, “Detection of modified measles and super-spreader using a real time reverse transcription pcr in the largest measles outbreak, yamagata, japan, 2017 in its elimination era,” *Epidemiology and Infection*, vol. 146, no. 14, pp. 1832–1839, Oct. 2018.
- [4] L. Warrener *et al.*, “Evaluation of a rapid diagnostic test for measles igm detection: Accuracy and the reliability of visual reading using sera from the measles surveillance programme in brazil, 2015,” *Epidemiology and Infection*, vol. 151, p. e151, 2023.
- [5] W. G. Madden *et al.*, “Deep neural networks for endemic measles dynamics: Comparative analysis and integration with mechanistic models,” *PLoS Computational Biology*, vol. 20, no. 11, p. e1012616, Nov. 2024.
- [6] S. A. Kujawski *et al.*, “Prediction of measles cases in us counties: A machine learning approach,” *Vaccine*, vol. 42, no. 26, p. 126289, Dec. 2024.
- [7] M. S. Y. Lau *et al.*, “Comparing and linking machine learning and semi-mechanistic models for the predictability of endemic measles dynamics,” *PLoS Computational Biology*, vol. 18, no. 9, p. e1010251, Sep. 2022.

- [8] R. Gyebi *et al.*, “Prediction of measles patients using machine learning classifiers: A comparative study,” *Bulletin of the National Research Centre*, vol. 47, p. 115, Jul. 2023.
- [9] B. Ru *et al.*, “Predicting measles outbreaks in the united states: Evaluation of machine learning approaches,” *JMIR Formative Research*, vol. 7, p. e42832, Apr. 2023.
- [10] P. A. Rota *et al.*, “Sequencing-based detection of measles in wastewater: Texas, january 2025,” *American Journal of Public Health*, vol. 115, no. 5, pp. 789–795, May 2025.
- [11] M. S. E. H. Hassan *et al.*, “Diagnostic approaches for measles virus: Methods, advances, and ongoing challenges,” *Pathogens*, vol. 14, no. 12, p. 1295, Dec. 2025.
- [12] A. V. S. L. N. S. Rao *et al.*, “Accurate measles rash detection via vision transformer fine-tuning,” May 2020.
- [13] A. M. Silva *et al.*, “A transparent hybrid deep learning framework to assess german measles disease in skin images,” in *Proceedings of the IEEE International Conference on Advanced Electrical, Electronic, Computers and Industrial Engineering (ICAEECI)*, Dec. 2023, pp. 1–6.
- [14] L. R. Santos *et al.*, “Detection of measles virus genotype a in a non-endemic wastewater setting: Insights from measles wastewater and environmental monitoring in canada’s capital region,” *Environmental Science & Technology Letters*, vol. 11, no. 9, pp. 910–916, Sep. 2024.